MGMT 405 – Managerial Economics

Teaching Note – Module 5: MR = MC

Preliminaries

This note explains how to maximize profits in the generic case of a firm facing a downward sloping, linear demand curve. Such a firm has a simple decision to make: choose a spot on the demand curve (i.e., a combination of price charged and quantity sold) such that its profits are maximized. This choice entails marginal decision making, picking *Q* to set marginal revenue equal to marginal cost: ***MR=MC***.

To set *MR=MC*, you need to know both *MR* and *MC*.

You should already know how to calculate *MR* (see Module 2 and the Teaching Note on Marginal Revenue). I will briefly repeat the steps starting from a specific demand function (from the lecture slides).

We start with the following demand:

$$Q = 1{,}600 - 4P$$

1) **Calculate inverse demand** (solve for P):

$$P = 400 - \frac{1}{4}Q$$

2) **Calculate total revenue** (multiply P(Q) by Q):

$$\mathrm{TR} = 400Q - \frac{1}{4}Q^2$$

3) **Compute marginal revenue from total revenue** (MR = dTR/dQ):

$$\mathrm{MR} = \frac{d\mathrm{TR}}{dQ} = 400 - \frac{1}{2}Q$$

Next, you need information on marginal cost. You obtain *MC* by taking the derivative of total cost (*TC*) with respect to quantity. Let's assume the following total cost: $\mathrm{TC} = 10{,}000 + 200Q$. Then:

$$\mathrm{MC} = \frac{d\mathrm{TC}}{dQ} = 200$$

Note that we're using the simplest case where marginal cost is constant.

The Final Step

Now we set *MR* equal to *MC*:

$$\begin{aligned} \mathrm{MR} &= \mathrm{MC} \\ \Leftrightarrow 400 - \frac{1}{2}Q &= 200 \\ \Leftrightarrow Q &= 400 \end{aligned}$$

So now we know that maximizing profits entails setting *Q*=400. What about the price *P*? Well, that's easy to find: our price-quantity combination must be located on the (inverse) demand curve (i.e., the one that we plot, with *P* on the y-axis). So from the (inverse) demand we derived earlier, we find:

$$P = 400 - \frac{1}{4}Q = 400 - 100 = \$300$$

Solution: Maximizing profits, here, entails setting **Q = 400 units** and **P = $300**.

Graphically: